

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Civil) (Credit System) Degree Examinations**

January 2023

20CV502 – DESIGN OF RCC STRUCTURAL ELEMENTS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) IS:456-2000 and SP 16 is permitted. Assume missing data suitably.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define and briefly explain (i) partial safety factor for loads (ii) effective depth (iii) lever arm.	5	L*2	1	1,8
	b) Explain difference between Working stress method and Limit state method.	5	L2	1	1,8
	c) Explain under reinforced, over reinforced and balanced section with strain diagram.	10	L2	1	1,12
2.	a) Explain Characteristic strength and Characteristic loads.	5	L2	1	1,8
	b) Derive an expression for compressive force with a neat sketch of stress strain block parameters of single reinforced RC beam.	5	L3	2	1,8
	c) A singly reinforced beam 250 mm X 500 mm in section is reinforced with 4 bars of 16 mm diameter with an effective cover of 50 mm. Effective span of the beam is 6 m. Assume M20 concrete and Fe 415 steel. Determine the central concentrated load 'P' that can be carried by the beam in addition to its self-weight.	10	L3	2	1,8
3.	a) A rectangular RCC beam is of 250 mm X 500 mm Overall size, with an effective cover of 50 mm on both tension and compression sides. It is reinforced with 4 bars of 20 mm bars on compression sides. Compute the steel on tension side with 20 mm bars and find the total moment of resistance of the section. Use M25 concrete and Fe 415 steel.	10	L3	2	1,8
	b) A RCC beam 250 mm wide and 450 mm deep, is reinforced with 3 numbers of 20 mm diameter bars of Fe 415, on tension side with an effective cover of 50 mm. If the shear reinforcement of 2 legged 8 mm stirrups at a spacing of 160 mm C/C is provided at a section. Determine the design strength. Assume M20 concrete.	10	L3	2	1,8
Unit – II					
4.	A rectangular beam is to be simply supported on supports of 230 mm width. The clear span of the beam is 6 m. The beam is to have width of 300 mm. The characteristic super imposed load is 12 kN/m. Using M20 concrete and Fe 415 steel design the beam.	20	L6	3	1,2,3
5.	A column of 400 mm X 400 mm is subjected to the factored loads $P_u = 1300$ kN, $M_{ux} = 190$ kN-m and $M_{uy} = 110$ kN-m. Design the reinforcement in the column, assuming M25 concrete and Fe 415 steel and effective cover of 60 mm. Assume it is a short column.	20	L6	4	1,2,3
6.	Design a Circular column of diameter 400 mm with helical reinforcement subjected to working load of 1200 kN. Use M25 concrete and Fe 415 steel. The column has unsupported length of 3m and its effective held in position at both ends, but not restrained against rotation.	20	L6	3	1,2,3

P.T.O.

Unit – III

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| 7. a) | Explain the procedure for the design of the cantilever slab. | 4 | L2 | 5 | 1,2,3 |
| b) | A hall has a clear dimension 3 m X 9 m with wall thickness 230 mm. The live load on the slab is 3 kN/m ² and a finish load of 1 kN/m ² . Using M20 grade of concrete and Fe 415 grade steel, design the slab. | 16 | L6 | 5 | 1,2,3 |
| 8. | Design a slab for a room of clear internal dimension 3 m X 5 m supported on walls of 300 mm thickness, with corners held down. Two adjacent edges of the slab are continuous and other two discontinuous. Live load on the slab is 3 kN/m ² . Assume floor finish of 1 kN/m ² . Use M20 concrete and Fe 415 steel. Sketch the details of reinforcements. | 20 | L6 | 5 | 1,2,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
